

Mahima Rudrapati

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Education

University of California, Davis

Master of Science in Computer Science, GPA: 3.94

Davis, California

Sep. 2024 – Jun 2026

University of Mumbai, India

Bachelor of Engineering in Computer Engineering (Minor in Artificial Intelligence), GPA: 3.85

Mumbai, India

Jul 2020 – May 2024

Experience

Teaching Assistant

University of California, Davis

Mar 2025 - Dec 2025

Davis, CA

- Taught 100+ students C++, data structures, and object-oriented programming through interactive code demonstrations.
- Led weekly office hours to resolve complex algorithmic and runtime issues, guiding students in software design and clean coding practices. Adapted course materials and assignments based on student performance analytics and feedback.

Software Engineer - Lead Frontend

Pilotcrew AI; Deployed Link: pilotcrew.ai

Jun 2025 - Sep 2025

Davis, CA

- Built an end-to-end LLM benchmarking interface using **React, TypeScript, and Tailwind CSS** to support automated and human-evaluated comparisons across 47 large language models in a cross-functional 5-member team.
- Optimized query handling through server-side rendering, reducing response latency to **under 500ms** for 10K+ records.
- Developed a parallel chunked file upload architecture with integrated validation, metadata extraction, and pre-processing workflows, enabling stable and consistent 1 GB dataset uploads in **under 75s**.
- Engineered reusable responsive UI components (real-time radial progress bars, data tables) and intuitive guiding tooltip features for continuous LLM performance tracking, enhancing interface interactivity and responsiveness.

Web Development Intern

Skrapnest; Deployed Link: skrapnest-prototype.vercel.app

Aug 2023 – Nov 2023

Mumbai, India

- Architected and deployed the full-stack prototype for a waste-management platform connecting over 20 scrap dealers and consumers, using **React, Firebase, JavaScript, and CSS** to deliver a functional MVP that secured \$5K in seed funding.
- Integrated **REST APIs** for real-time order management, enabling role-based operations and data synchronization.
- Implemented Firebase OTP authentication with validation pipelines, strengthening secure access and data integrity.

Projects

Restaurant Recommendation System using ML Algorithms and Web Scraping [First Author - IEEE Xplore Publication]

- Led a team of 4 to build a scalable ML-driven recommendation platform with **React, PostgreSQL, Flask, and Python**.
- Designed and deployed RESTful APIs with ERDs/DFDs to perform CRUD operations **over 10K+** restaurants and reviews.
- Implemented real-time scraping pipelines using **Selenium and Botasaurus**, achieving **300** record updates in **90s**.
- Applied NLP pipelines for opinion mining, user reliability and feature extraction to enrich user sentiment modeling.
- Integrated collaborative and content-based filtering using tag-profiled embeddings to improve accuracy.

Autonomous Vision System for Space Robotics (Computer Vision)

- Built a **Python**-based visual localization pipeline using AprilTags and RGB-D camera SDKs for an autonomous robot vision system designed to guide precise plant growth in space, reaching **under sub-0.2 px** reprojection error after calibration.
- Processed 3D point clouds using **OpenCV, Open3D, and C++**, reaching **over 95%** geometric estimation alignment.
- Deployed Meta SAM and NVIDIA FoundationPose models for segmenting growing plants, enabling dynamic perception.

Data Visualization of Green House Gas Emissions in America (Data Science, UI&UX Design)

- Built an interactive **D3.js** and React-based data visualization dashboard that analyzed over 500K CO/kg emission records across multiple industrial sectors to identify and compare key emission contributors.
- Engineered drill-down visual analytics (hierarchical bubbles, stacked bars, pie charts) to surface insights.

Technologies

Languages & Databases: Python, TypeScript/JavaScript, HTML/CSS, C/C++, SQL, PostgreSQL, MongoDB

Frameworks: React, React Native, Node.js, Flask, FastAPI, Redux Toolkit, Tailwind CSS, Selenium

Tools: Git/GitHub, Docker, Firebase, Linux/Unix, Jira, Figma